LOT PAGER ??? 05-FIRMWARE

LOT® PAGER — Firmware Specification

Target: ESP32-S3, ESP-IDF 5.x (C/C++), FreeRTOS. License: LOT® proprietary.

1. Modules

Module	Responsibility
`net`	Wi-Fi STA, TLS 1.2+, reconnect with backoff
`pair`	One-time pairing code → device token (06 §2); token in NVS, encrypted (flash encryption on)
`api`	Poll/stream notifications, post events, upload frames
`ui`	Notification card, status bar (battery, Wi-Fi), pager "ack" animation
`input`	Button debounce, short/long press
`sense`	Sample BME688/BMI270/VEML at set intervals; batch & compress
`cam`	Capture JPEG on demand/trigger; never streams continuously
`pwr`	Charge state, fuel gauge, deep sleep
`ota`	Signed OTA (dual partition, rollback)
`log`	Ring buffer in RAM + NVS; upload on connect

2. State machine

```
BOOT → (no token) PAIRING → ONLINE ⇄ SLEEP
ONLINE: poll (30 s active / 5 min idle) → NOTIFY → [button] → ACK → ONLINE
CHARGING: screen shows level; radio throttled
```

3. Notification behavior (pager-like)

- Payload ≤ 240 bytes UTF-8: `{id, text, level, ts, ttl}`.
- Display: text up to 3 lines, auto-fit; buzz pattern by level (info 1 short, nudge 2 short, alert long).
- Stays on screen until ack or ttl. Backlog: keep last 8, newest first.
- Quiet hours honored from server prefs (device never decides to be silent by itself except low battery <5 %).

4. Button "Copy"

- Short press → `POST /api/logs` with `event:"device.copy"`, text = the notification text (see 06 §4). Screen flashes “Copied”, then clears the card.
- Long press (1.5 s) → request a camera capture.
- If offline, event is queued (max 100) and sent on reconnect with original timestamp.

5. Power budget (indicative, 200 mAh)

State	Current	Share of time	Average
Deep sleep + RTC	20 µA	70 %	14 µA
Wi-Fi poll (light sleep between)	80 mA for 2 s / 5 min	0.7 %	~0.5 mA
Screen on (user viewing)	60 mA	1 %	0.6 mA
Camera shot	150 mA for 3 s	rare	negligible

→ ~1.2 mA average ⇒ **~7 days** per charge. Wake on BMI270 pickup extends interactivity.

6. Camera policy

Capture only on explicit trigger (long press or server request that shows an LED + on-screen indicator). Frames are JPEG ≤ 640×480, encrypted in transit, deleted on device after upload. Every capture produces a Log entry.

7. Security

Secure boot v2, flash encryption, TLS cert pinning to the LOT API host, per-device token revocable server-side, no inbound ports.

8. Testing

Host-side unit tests for parsing/queue (Unity); HIL smoke test using pager-sim.mjs as server; 72 h soak.